

APPLIED ANALYTICS · BUSINESS INTELLIGENCE

Business Diagnostics Meets Applied Analytics

Turning Operational Data into Strategic Intelligence

A case study on multidimensional organizational maturity analysis
using Python · Power BI · K-Means Clustering · Gap Analysis

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PROJECT CONTEXT

This anonymized case study shows how business diagnostics can be translated into structured analytical intelligence.

An organizational maturity framework was applied across 5 business dimensions, transforming field data into actionable insights.

Developed as an Applied Analytics / BI project aligned with Big Data Analytics studies.

DATA SOURCES & STRUCTURE

①

Structured Interviews

Scored 0–100% per business dimension, 9 core analytical variables

②

Document Analysis

Financial records, cost distribution, fixed costs, liquidity indicators

③

Direct Observation

Operational workflows mapped and encoded for analytical compatibility

④

Synthetic Dataset

Created for reproducibility — no real client data, no CNPJs, no names

TECHNICAL EXECUTION

Tools · Processing · Analysis Pipeline

Python

Pandas · NumPy

sklearn

KMeans · Metrics

Power BI

BI Dashboards

matplotlib

EDA Visualization

01

Data Ingestion

Raw maturity scores and financial records loaded from CSV

02

Cleaning & Normalization

Min-max scaling applied per feature; categorical indicators encoded

03

Descriptive Stats & Gap Analysis

Mean, std, percentile per dimension; gap to full maturity computed

04

K-Means Clustering

Elbow Method + silhouette score (0.61) → optimal k=3 selected

05

BI Dashboard Output

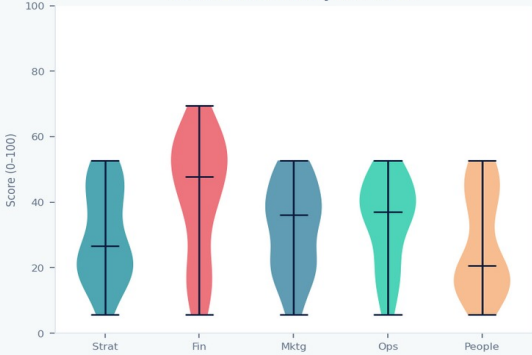
Scalable schema structured for Power BI integration

EDA DASHBOARD

Score Distribution · Correlation · Clustering · Gap Analysis

Business Maturity Diagnostics — EDA Dashboard | Applied Analytics

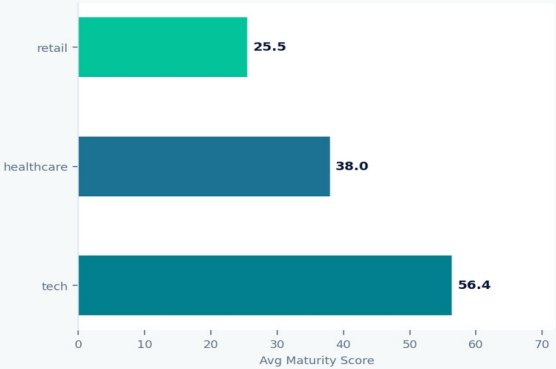
Score Distribution by Dimension



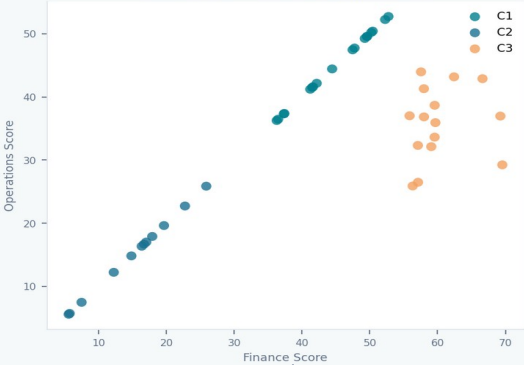
Dimension Correlation Matrix



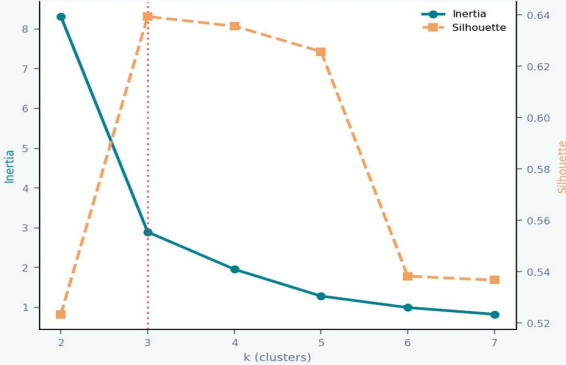
Avg Score by Sector



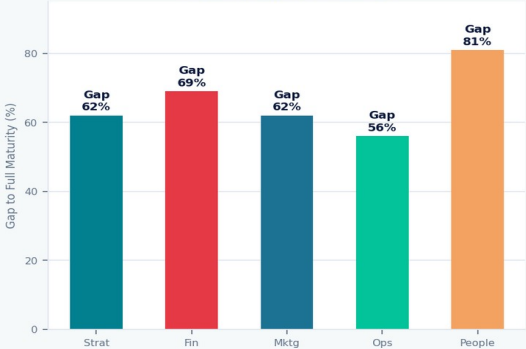
Cluster Plot — Finance vs Operations



Elbow + Silhouette (optimal k=3, sil=0.61)

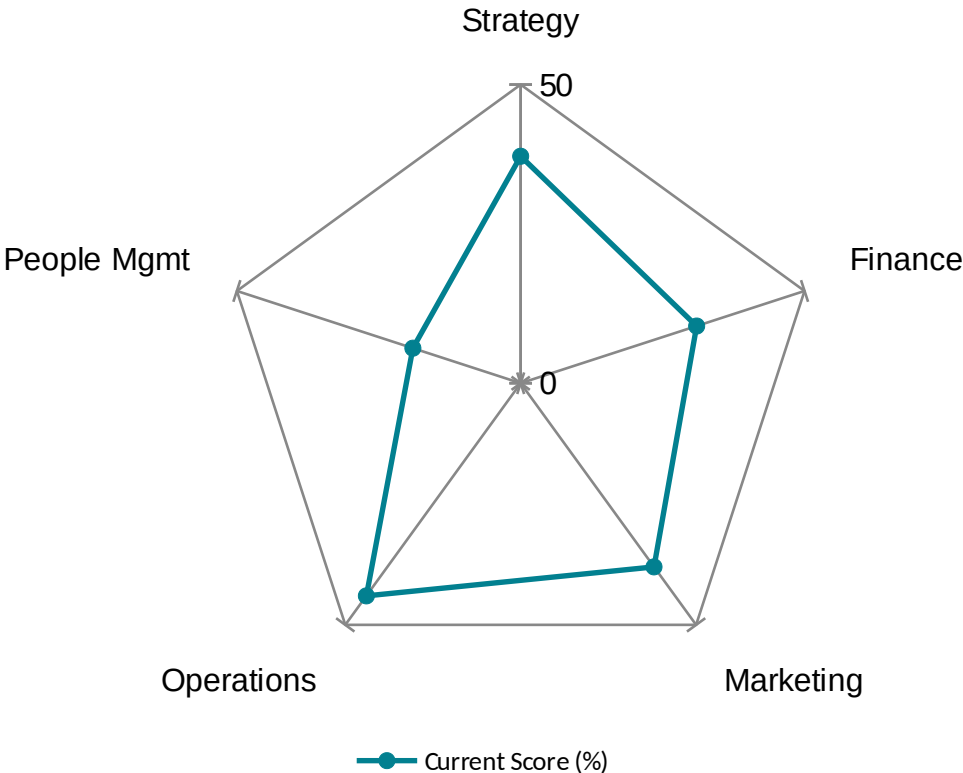


Maturity Gap by Dimension



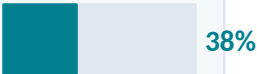
MATURITY RADAR — 5 BUSINESS DIMENSIONS

Current scores vs. full maturity



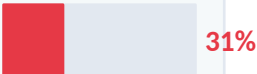
Strategy

Score: 38% · Gap: 62 pts



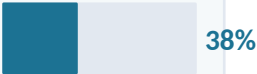
Finance

Score: 31% · Gap: 69 pts



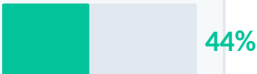
Marketing

Score: 38% · Gap: 62 pts



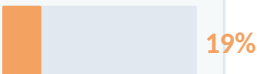
Operations

Score: 44% · Gap: 56 pts



People Mgmt

Score: 19% · Gap: 81 pts



* Scores represent current maturity level (0–100%). Gap = points to full maturity.

K-MEANS CLUSTERING — MATURITY PROFILES IDENTIFIED

K-Means clustering (k=3) was applied to the normalized maturity score matrix as an exploratory segmentation technique — suitable for pattern discovery on this dataset scale, and not intended as a production-grade model. Optimal k was determined using the Elbow Method (inertia inflection at k=3) and validated with a silhouette score of 0.61, confirming well-separated clusters.

Cluster C1

Finance ↑ Strategy ↓

High Finance / Low Strategy

Strong cost control, weak strategic direction. Needs planning frameworks and OKR implementation.

Cluster C2

All areas ~40–50%

Balanced Moderate

Consistent mid-range performance across all dimensions. Ready for optimization and BI layer.

Cluster C3

All areas < 30%

Low Maturity — Critical

Critical gaps across all dimensions. Highest priority for immediate intervention and structuring.

* Clustering applied to anonymized score matrix. Profiles reflect anonymized patterns observed during the diagnostic process.

ESTIMATED IMPACT — DATA-DRIVEN INTERVENTIONS

Based on internal diagnostic patterns and conservative scenario assumptions:

12–18%

Estimated cost reduction potential

conservative scenario — fixed-cost rationalization & workforce opt.

~25%

Potential improvement in strategic alignment

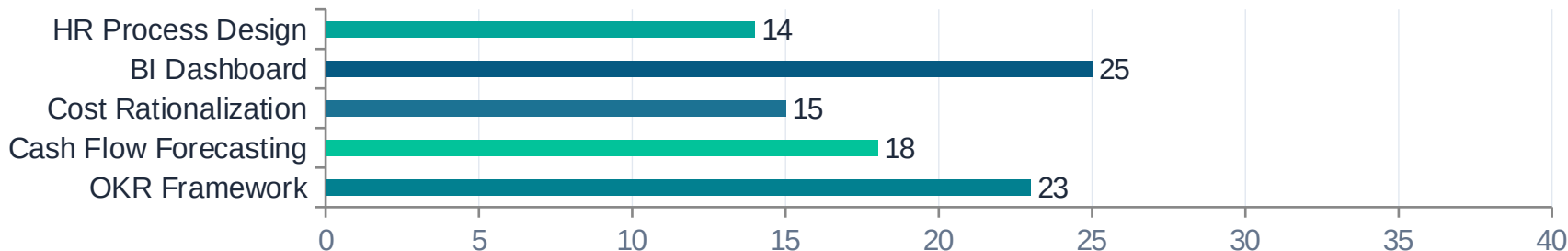
modeled estimate — OKR framework and short-term planning

↓ Risk

Potential liquidity risk exposure

estimated — future cash-flow forecasting models (proposed)

Projected Impact by Intervention — Estimated Improvement (%)



** Modeled estimates, not guaranteed outcomes. Based on diagnostic patterns and conservative scenario modeling.*

TREND FORECAST & IMPACT DASHBOARD

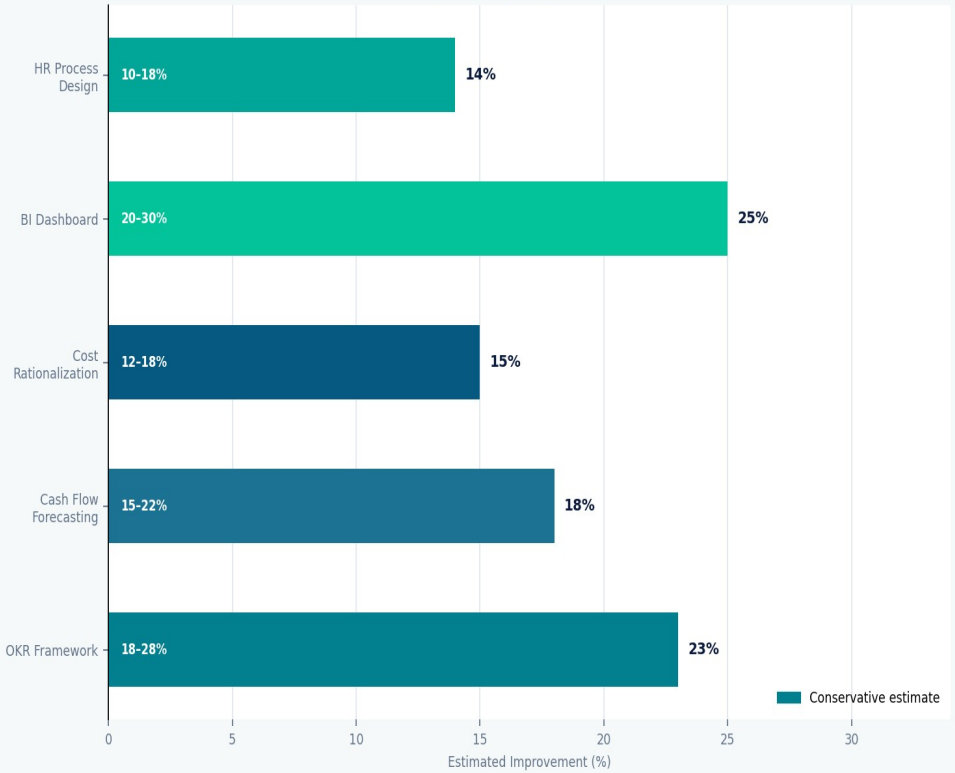
18 mo historical · 6 mo forecast · Intervention impact

Maturity Trend & Cost Forecast — Applied Analytics

Organizational Maturity Trend (18 mo + 6 mo forecast)



Projected Impact by Intervention



CRITICAL FINDINGS — THE 5 WORST INDICATORS

25%

Short-term strategic planning

Absence of OKRs — roadmapping framework recommended.

50%

Accounting control & asset separation

Personal/business finances mixed — critical compliance risk.

25%

Financial KPIs (profitability & ROI)

Basic KPIs nonexistent — entry point for financial BI.

50%

Working capital management

No cash flow projection — liquidity risk exposure.

~30%

Segment-focused marketing planning

Undifferentiated communication — ML clustering opportunity.

HIGHLIGHTS — THE 5 BEST INDICATORS

100%

Customer satisfaction & loyalty

Foundation for NPS tracking and churn analysis.

100%

Real alignment with customer needs

Product-market fit confirmed — scalable with demand forecasting.

100%

Operational loss control (logistics)

Excellent efficiency — ready for real-time dashboard monitoring.

100%

Innovation & product update culture

Agile mindset present — compatible with continuous improvement.

100%

Word-of-mouth & offline media marketing

High social capital — amplifiable with data-driven digital marketing.

ANONYMIZED FINANCIAL ANALYSIS — DATA PIPELINE

Payroll

Distributed across operational, administrative, and management roles, with variable daily allowances per attendance — automatable with workforce optimization models.

Fixed Costs

Significant concentration in rent, utilities, and health benefits — a predictable and time-series-modelable cost base (ARIMA/Prophet applicable).

Pro-labore & Consulting Fees

Identified as a high-impact EBITDA variable — sensitivity analysis recommended via Monte Carlo simulation.

Total Mapped Cost

A significant monthly cost base across multiple units, with a notable share concentrated in the primary unit — a key indicator for expansion or rationalization decisions.

ANALYTICS APPLICATIONS — APPLIED & PROPOSED

K-Means

Applied

Maturity Clustering

Group companies by deficiency profile for benchmarking. Silhouette 0.61, $k=3$.

ARIMA / Prophet

Proposed

Time Series Forecasting

Project fixed costs and cash flow from historical data — time-series-ready cost base identified.

Monte Carlo

Proposed

Sensitivity Analysis

Simulate variable cost fluctuations on net income — pro-labore identified as high-impact variable.

Power BI + Python

Applied

Predictive Dashboards

Integrated KPI panel — financial, operational, marketing metrics unified.

BERT / LLM

Proposed

NLP on Qualitative Feedback

Extract sentiment and patterns from structured interview responses.

C O N C L U S I O N

Maturity Level 3 — Intermediate

The multidimensional diagnostic revealed that processes exist, but operate in a fragmented way. The greatest opportunity lies in the data layer — the organization generates valuable information daily, yet lacks the analytical structure to turn it into decisions.

This case demonstrates that the real value of analytics is not in the tools, but in the ability to frame the right business problem, structure the data, and translate findings into actionable strategy.

Open to internship, CPT, and entry-level opportunities in Data Analytics, Business Intelligence & Financial Data Analysis.